```
!pip install kagglehub pandas numpy matplotlib seaborn scikit-learn plotly
```

```
Requirement already satisfied: kagglehub in /usr/local/lib/python3.13/dist-packages (1.0.3)
Requirement already satisfied: pandas in /usr/local/lib/python3.13/dist-packages (2.2.3)
Requirement already satisfied: numpy in /usr/local/lib/python3.13/dist-packages (2.1.3)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.13/dist-packages (3.10.0)
Requirement already satisfied: seaborn in /usr/local/lib/python3.13/dist-packages (0.13.2)
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.13/dist-packages (1.6.1)
Requirement already satisfied: plotly in /usr/local/lib/python3.13/dist-packages (5.24.1)
Requirement already satisfied: certifi>=2026.10.19 in /usr/local/lib/python3.13/dist-packages (from kagglehub) (2026.10.
Requirement already satisfied: kagglesdk<1.0,>=0.1.38 in /usr/local/lib/python3.13/dist-packages (from kagglehub) (0.1.3
Requirement already satisfied: packaging in /usr/local/lib/python3.13/dist-packages (from kagglehub) (26.3)
Requirement already satisfied: pyyaml in /usr/local/lib/python3.13/dist-packages (from kagglehub) (6.0.3)
Requirement already satisfied: requests in /usr/local/lib/python3.13/dist-packages (from kagglehub) (2.32.4)
Requirement already satisfied: tqdm in /usr/local/lib/python3.13/dist-packages (from kagglehub) (4.67.3)
Requirement already satisfied: protobuf in /usr/local/lib/python3.13/dist-packages (from kagglesdk<1.0,>=0.1.38->kaggleh
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.13/dist-packages (from pandas) (2.9.0.po
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.13/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.13/dist-packages (from pandas) (2026.5)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.13/dist-packages (from matplotlib) (1.4.0)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.13/dist-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.13/dist-packages (from matplotlib) (4.67.0)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.13/dist-packages (from matplotlib) (1.5.1)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.13/dist-packages (from matplotlib) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.13/dist-packages (from matplotlib) (3.3.3)
Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.13/dist-packages (from scikit-learn) (1.16.3)
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.13/dist-packages (from scikit-learn) (1.6.0)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.13/dist-packages (from scikit-learn) (3.7.
Requirement already satisfied: tenacity>=6.2.0 in /usr/local/lib/python3.13/dist-packages (from plotly) (9.1.4)
Requirement already satisfied: cloudpickle>=3.0 in /usr/local/lib/python3.13/dist-packages (from joblib>=1.2.0->scikit-l
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.13/dist-packages (from python-dateutil>=2.8.2->pandas)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.13/dist-packages (from requests->kaggl
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.13/dist-packages (from requests->kagglehub) (3.20)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.13/dist-packages (from requests->kagglehub)
```

```
import kagglehub

path = kagglehub.dataset_download(
    "shuyangli94/food-com-recipes-and-user-interactions"
)

print("Dataset location:", path)
```

```
Downloading to /root/.cache/kagglehub/datasets/shuyangli94/food-com-recipes-and-user-interactions/2.archive...
100%|██████████| 267M/267M [00:02<00:00, 96.3MB/s]Extracting files...

Dataset location: /root/.cache/kagglehub/datasets/shuyangli94/food-com-recipes-and-user-interactions/versions/2
```

```
import os

print(os.listdir(path))
```

```
['interactions_validation.csv', 'interactions_test.csv', 'RAW_interactions.csv', 'interactions_train.csv', 'RAW_recipes.
```

```
import pandas as pd
import os

recipes = pd.read_csv(
    os.path.join(path, "RAW_recipes.csv")
)

interactions = pd.read_csv(
    os.path.join(path, "RAW_interactions.csv")
)

print("Recipes shape:", recipes.shape)
print("Interactions shape:", interactions.shape)
```

```
Recipes shape: (231637, 12)
Interactions shape: (1132367, 5)
```

```
print(recipes.columns)
print(interactions.columns)
```

```
Index(['name', 'id', 'minutes', 'contributor_id', 'submitted', 'tags',
       'nutrition', 'n_steps', 'steps', 'description', 'ingredients',
       'n_ingredients'],
      dtype='object')
Index(['user_id', 'recipe_id', 'date', 'rating', 'review'], dtype='object')
```

```
recipes.head()
```

	name	id	minutes	contributor_id	submitted	tags	nutrition	n_steps	steps	description	ingredi
0	arriba baked winter squash mexican style	137739	55	47892	2005-09-16	['60-minutes-or-less', 'time-to-make', 'course...	[51.5, 0.0, 13.0, 0.0, 2.0, 0.0, 4.0]	11	['make a choice and proceed with recipe', 'dep...	autumn is my favorite time of year to cook! th...	['v squ 'me seaso 'mix
1	a bit different breakfast pizza	31490	30	26278	2002-06-17	['30-minutes-or-less', 'time-to-make', 'course...	[173.4, 18.0, 0.0, 17.0, 22.0, 35.0, 1.0]	9	['preheat oven to 425 degrees f', 'press dough...	this recipe calls for the crust to be prebaked...	['prep pizza c 'sau patty', '(
2	all in the kitchen chili	112140	130	196586	2005-02-25	['time-to-make', 'course', 'preparation', 'mai...	[269.8, 22.0, 32.0, 48.0, 39.0, 27.0, 5.0]	6	['brown ground beef in large pot', 'add choppe...	this modified version of 'mom's' chili was a h...	['gr beef', 'y on ' tom
3	alouette potatoes	59389	45	68585	2003-04-14	['60-minutes-or-less', 'time-to-make', 'course...	[368.1, 17.0, 10.0, 2.0, 14.0, 8.0, 20.0]	11	['place potatoes in a large pot of lightly sal...	this is a super easy, great tasting, make ahea...	['sprea cheese garli herbs
4	amish tomato ketchup for canning	44061	190	41706	2002-10-25	['weeknight', 'time-to-make', 'course', 'main-...	[352.9, 1.0, 337.0, 23.0, 3.0, 0.0, 28.0]	5	['mix all ingredients& boil for 2 1 / 2 hours ...	my dh's amish mother raised him on this recipe...	['to juice', ' vin 'su

```
interactions.head()
```

	user_id	recipe_id	date	rating	review
0	38094	40893	2003-02-17	4	Great with a salad. Cooked on top of stove for...
1	1293707	40893	2011-12-21	5	So simple, so delicious! Great for chilly fall...
2	8937	44394	2002-12-01	4	This worked very well and is EASY. I used not...
3	126440	85009	2010-02-27	5	I made the Mexican topping and took it to bunk...
4	57222	85009	2011-10-01	5	Made the cheddar bacon topping, adding a sprin...

```
recipes.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 231637 entries, 0 to 231636
Data columns (total 12 columns):
 #   Column          Non-Null Count   Dtype 
---  ------          --------------   ----- 
 0   name            231636 non-null  object
 1   id              231637 non-null  int64 
 2   minutes         231637 non-null  int64 
 3   contributor_id  231637 non-null  int64 
 4   submitted       231637 non-null  object
 5   tags            231637 non-null  object
 6   nutrition       231637 non-null  object
 7   n_steps         231637 non-null  int64 
 8   steps           231637 non-null  object
 9   description     226658 non-null  object
 10  ingredients     231637 non-null  object
 11  n_ingredients   231637 non-null  int64 
dtypes: int64(5), object(7)
memory usage: 21.2+ MB
```

```
recipes.isnull().sum()
```

	0
name	1
id	0
minutes	0
contributor_id	0
submitted	0
tags	0
nutrition	0
n_steps	0
steps	0
description	4979
ingredients	0
n_ingredients	0

dtype: int64

```
recipes.duplicated().sum()
```

```
np.int64(0)
```

```
recipes["id"].duplicated().sum()
```

```
np.int64(0)
```

```
interactions["user_id"].duplicated().sum()
```

```
np.int64(905797)
```

```
print("Missing values in recipes:")
print(recipes.isnull().sum())

print("\nMissing values in interactions:")
print(interactions.isnull().sum())
```

```
Missing values in recipes:
name                1
id                  0
minutes             0
contributor_id      0
submitted           0
tags                0
nutrition           0
n_steps             0
steps               0
description      4979
ingredients         0
n_ingredients       0
dtype: int64

Missing values in interactions:
user_id        0
recipe_id      0
date           0
rating         0
review       169
dtype: int64
```

```
print("Recipes:", recipes.shape)
print("Interactions:", interactions.shape)
```

```
Recipes: (231637, 12)
Interactions: (1132367, 5)
```

```
recipes.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 231637 entries, 0 to 231636
Data columns (total 12 columns):
 #   Column          Non-Null Count   Dtype
---  ------          --------------   -----
 0   name            231636 non-null  object
```

```
 1   id              231637 non-null  int64
 2   minutes         231637 non-null  int64
 3   contributor_id  231637 non-null  int64
 4   submitted       231637 non-null  object
 5   tags            231637 non-null  object
 6   nutrition       231637 non-null  object
 7   n_steps         231637 non-null  int64
 8   steps           231637 non-null  object
 9   description     226658 non-null  object
 10  ingredients     231637 non-null  object
 11  n_ingredients   231637 non-null  int64
dtypes: int64(5), object(7)
memory usage: 21.2+ MB
```

```
interactions.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1132367 entries, 0 to 1132366
Data columns (total 5 columns):
 #   Column     Non-Null Count    Dtype
---  ------     --------------    -----
 0   user_id    1132367 non-null  int64
 1   recipe_id  1132367 non-null  int64
 2   date       1132367 non-null  object
 3   rating     1132367 non-null  int64
 4   review     1132198 non-null  object
dtypes: int64(3), object(2)
memory usage: 43.2+ MB
```

```
display(recipes.head(5))
```

	name	id	minutes	contributor_id	submitted	tags	nutrition	n_steps	steps	description	ingredi
0	arriba baked winter squash mexican style	137739	55	47892	2005-09-16	['60-minutes-or-less', 'time-to-make', 'course...	[51.5, 0.0, 13.0, 0.0, 2.0, 0.0, 4.0]	11	['make a choice and proceed with recipe', 'dep...	autumn is my favorite time of year to cook! th...	['v squ 'me seaso 'mix
1	a bit different breakfast pizza	31490	30	26278	2002-06-17	['30-minutes-or-less', 'time-to-make', 'course...	[173.4, 18.0, 0.0, 17.0, 22.0, 35.0, 1.0]	9	['preheat oven to 425 degrees f', 'press dough...	this recipe calls for the crust to be prebaked...	['prep pizza c 'sau patty', '(
2	all in the kitchen chili	112140	130	196586	2005-02-25	['time-to-make', 'course', 'preparation', 'mai...	[269.8, 22.0, 32.0, 48.0, 39.0, 27.0, 5.0]	6	['brown ground beef in large pot', 'add choppe...	this modified version of 'mom's' chili was a h...	['gr beef', 'y on ' tom
3	alouette potatoes	59389	45	68585	2003-04-14	['60-minutes-or-less', 'time-to-make', 'course...	[368.1, 17.0, 10.0, 2.0, 14.0, 8.0, 20.0]	11	['place potatoes in a large pot of lightly sal...	this is a super easy, great tasting, make ahea...	['sprea cheese garli herbs
4	amish tomato ketchup for canning	44061	190	41706	2002-10-25	['weeknight', 'time-to-make', 'course', 'main-...	[352.9, 1.0, 337.0, 23.0, 3.0, 0.0, 28.0]	5	['mix all ingredients& boil for 2 1 / 2 hours ...	my dh's amish mother raised him on this recipe...	['to juice', ' vin 'su

```
display(interactions.head(5))
```

	user_id	recipe_id	date	rating	review
0	38094	40893	2003-02-17	4	Great with a salad. Cooked on top of stove for...
1	1293707	40893	2011-12-21	5	So simple, so delicious! Great for chilly fall...
2	8937	44394	2002-12-01	4	This worked very well and is EASY. I used not...
3	126440	85009	2010-02-27	5	I made the Mexican topping and took it to bunk...
4	57222	85009	2011-10-01	5	Made the cheddar bacon topping, adding a sprin...

```
print("Recipe columns:")
print(recipes.columns.tolist())

print("\nInteraction columns:")
print(interactions.columns.tolist())
```

```
Recipe columns:
['name', 'id', 'minutes', 'contributor_id', 'submitted', 'tags', 'nutrition', 'n_steps', 'steps', 'description', 'ingred

Interaction columns:
['user_id', 'recipe_id', 'date', 'rating', 'review']
```

```
recipes.describe()
```

	id	minutes	contributor_id	n_steps	n_ingredients
count	231637.000000	2.316370e+05	2.316370e+05	231637.000000	231637.000000
mean	222014.708984	9.398546e+03	5.534885e+06	9.765499	9.051153
std	141206.635626	4.461963e+06	9.979141e+07	5.995128	3.734796
min	38.000000	0.000000e+00	2.700000e+01	0.000000	1.000000
25%	99944.000000	2.000000e+01	5.690500e+04	6.000000	6.000000
50%	207249.000000	4.000000e+01	1.736140e+05	9.000000	9.000000
75%	333816.000000	6.500000e+01	3.982750e+05	12.000000	11.000000
max	537716.000000	2.147484e+09	2.002290e+09	145.000000	43.000000

```
interactions.describe()
```

	user_id	recipe_id	rating
count	1.132367e+06	1.132367e+06	1.132367e+06
mean	1.384291e+08	1.608972e+05	4.411016e+00
std	5.014269e+08	1.303987e+05	1.264752e+00
min	1.533000e+03	3.800000e+01	0.000000e+00
25%	1.354700e+05	5.425700e+04	4.000000e+00
50%	3.309370e+05	1.205470e+05	5.000000e+00
75%	8.045500e+05	2.438520e+05	5.000000e+00
max	2.002373e+09	5.377160e+05	5.000000e+00

Exploratory Data Analysis

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_theme(style="whitegrid")
plt.rcParams["figure.figsize"] = (10, 5)

print("Libraries imported successfully!")
```

```
Libraries imported successfully!
```

```
print("Total recipes:", len(recipes))
print("Total columns:", len(recipes.columns))

display(recipes.head())
```

```
Total recipes: 231637
Total columns: 12
```

	name	id	minutes	contributor_id	submitted	tags	nutrition	n_steps	steps	description	ingredi
0	arriba baked winter squash mexican style	137739	55	47892	2005-09-16	['60-minutes-or-less', 'time-to-make', 'course...	[51.5, 0.0, 13.0, 0.0, 2.0, 0.0, 4.0]	11	['make a choice and proceed with recipe', 'dep...	autumn is my favorite time of year to cook! th...	['v squ 'me seaso 'mix
1	a bit different breakfast pizza	31490	30	26278	2002-06-17	['30-minutes-or-less', 'time-to-make', 'course...	[173.4, 18.0, 0.0, 17.0, 22.0, 35.0, 1.0]	9	['preheat oven to 425 degrees f', 'press dough...	this recipe calls for the crust to be prebaked...	['prep pizza 'sau patty', '
2	all in the kitchen chili	112140	130	196586	2005-02-25	['time-to-make', 'course', 'preparation', 'mai...	[269.8, 22.0, 32.0, 48.0, 39.0, 27.0, 5.0]	6	['brown ground beef in large pot', 'add choppe...	this modified version of 'mom's' chili was a h...	['gr beef', 'y on ' tom
3	alouette potatoes	59389	45	68585	2003-04-14	['60-minutes-or-less', 'time-to-make', 'course...	[368.1, 17.0, 10.0, 2.0, 14.0, 8.0, 20.0]	11	['place potatoes in a large pot of lightly sal...	this is a super easy, great tasting, make ahea...	['sprea cheese garli herbs
4	amish tomato ketchup for canning	44061	190	41706	2002-10-25	['weeknight', 'time-to-make', 'course', 'main-...	[352.9, 1.0, 337.0, 23.0, 3.0, 0.0, 28.0]	5	['mix all ingredients& boil for 2 1 / 2 hours ...	my dh's amish mother raised him on this recipe...	['to juice', ' vin 'su

```
recipes[[
    "name",
    "minutes",
    "contributor_id",
    "submitted",
    "tags",
    "nutrition",
    "n_ingredients"
]].head(10)
```

	name	minutes	contributor_id	submitted	tags	nutrition	n_ingredients
0	arriba baked winter squash mexican style	55	47892	2005-09-16	['60-minutes-or-less', 'time-to-make', 'course...	[51.5, 0.0, 13.0, 0.0, 2.0, 0.0, 4.0]	7
1	a bit different breakfast pizza	30	26278	2002-06-17	['30-minutes-or-less', 'time-to-make', 'course...	[173.4, 18.0, 0.0, 17.0, 22.0, 35.0, 1.0]	6
2	all in the kitchen chili	130	196586	2005-02-25	['time-to-make', 'course', 'preparation', 'mai...	[269.8, 22.0, 32.0, 48.0, 39.0, 27.0, 5.0]	13
3	alouette potatoes	45	68585	2003-04-14	['60-minutes-or-less', 'time-to-make', 'course...	[368.1, 17.0, 10.0, 2.0, 14.0, 8.0, 20.0]	11
4	amish tomato ketchup for canning	190	41706	2002-10-25	['weeknight', 'time-to-make', 'course', 'main-...	[352.9, 1.0, 337.0, 23.0, 3.0, 0.0, 28.0]	8
5	apple a day milk shake	0	1533	1999-12-06	['15-minutes-or-less', 'time-to-make', 'course...	[160.2, 10.0, 55.0, 3.0, 9.0, 20.0, 7.0]	4
6	aww marinated olives	15	21730	2002-04-14	['15-minutes-or-less', 'time-to-make', 'course...	[380.7, 53.0, 7.0, 24.0, 6.0, 24.0, 6.0]	9
					['weeknight', 'time-to-		

```
print(recipes["minutes"].describe())
```

```
count    2.316370e+05
mean     9.398546e+03
std      4.461963e+06
min      0.000000e+00
25%      2.000000e+01
```

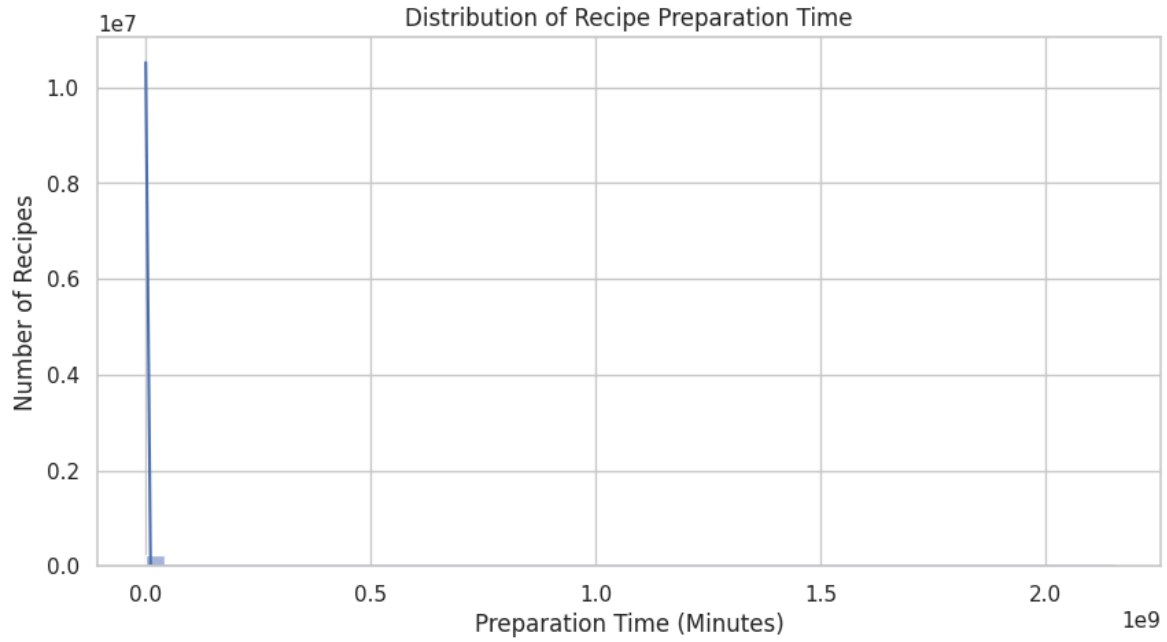
```
50%      4.000000e+01
75%      6.500000e+01
max      2.147484e+09
Name: minutes, dtype: float64
```

```
plt.figure(figsize=(10, 5))

sns.histplot(recipes["minutes"], bins=50, kde=True)

plt.title("Distribution of Recipe Preparation Time")
plt.xlabel("Preparation Time (Minutes)")
plt.ylabel("Number of Recipes")

plt.show()
```




```
recipes.nlargest(10, "minutes")[["name", "minutes"]]
```

	name	minutes
144074	no bake granola balls	2147483647
109624	how to preserve a husband	1051200
106563	homemade fruit liquers	288000
38116	celtic druid s honey mead meade metheglin	259260
106963	homemade vanilla	259205
155044	peach brandy	216015
5497	angelica liqueur	201610
177906	rumtopf	187200
170999	raspberry liqueur	172810
53721	cinnamon extract	172800

```
print(recipes["n_ingredients"].describe())
```

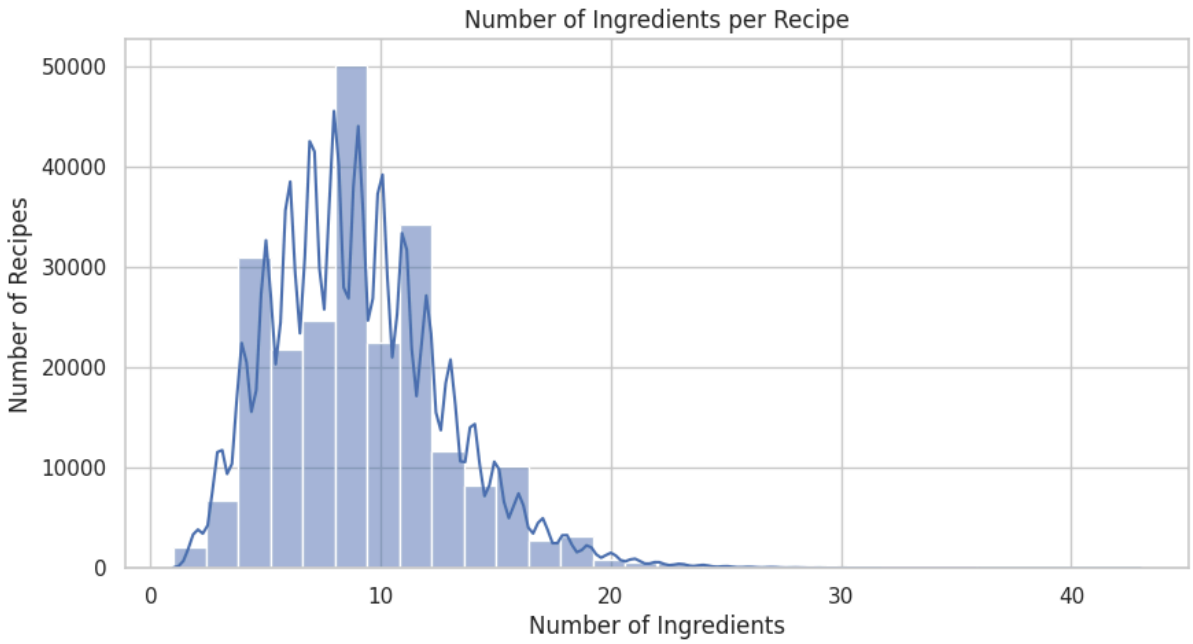
```
count    231637.000000
mean          9.051153
std           3.734796
min           1.000000
25%           6.000000
50%           9.000000
75%          11.000000
max          43.000000
Name: n_ingredients, dtype: float64
```

```
plt.figure(figsize=(10, 5))

sns.histplot(recipes["n_ingredients"], bins=30, kde=True)
```

```
plt.title("Number of Ingredients per Recipe")
plt.xlabel("Number of Ingredients")
plt.ylabel("Number of Recipes")

plt.show()
```




```
recipes.nlargest(10, "n_ingredients")[["name", "n_ingredients"]]
```

	name	n_ingredients
2162	a to z bread	43
139659	morton thompson turkey	40
230245	z best two faced jerk	40
116744	kato s garage style tomato sauce meatba...	39
18352	basic fried rice with variations	38
119626	land and sea white meat version of surf and turf	37
203614	sunday supper taco salad bowls tacos	37
71669	dhansak with brown rice	36
23572	bill s calzones	35
55542	coconut chicken fingers with mango rum salsa	35

```
from collections import Counter
import ast

ingredient_counter = Counter()

for ingredients in recipes["ingredients"].dropna():
    try:
        ingredient_list = ast.literal_eval(ingredients)

        for ingredient in ingredient_list:
            ingredient_counter[ingredient.lower().strip()] += 1

    except (ValueError, SyntaxError):
        continue

top_20_ingredients = ingredient_counter.most_common(20)

ingredient_df = pd.DataFrame(
    top_20_ingredients,
    columns=["Ingredient", "Frequency"]
)

display(ingredient_df)
```

	Ingredient	Frequency
0	salt	85746
1	butter	54975
2	sugar	44535
3	onion	39065
4	water	34914
5	eggs	33761
6	olive oil	32822
7	flour	26266
8	milk	25786
9	garlic cloves	25748
10	pepper	22319
11	brown sugar	18655
12	garlic	18087
13	all-purpose flour	17659
14	baking powder	17504
15	egg	17304
16	salt and pepper	15415
17	parmesan cheese	14807
18	lemon juice	14233
19	baking soda	14099

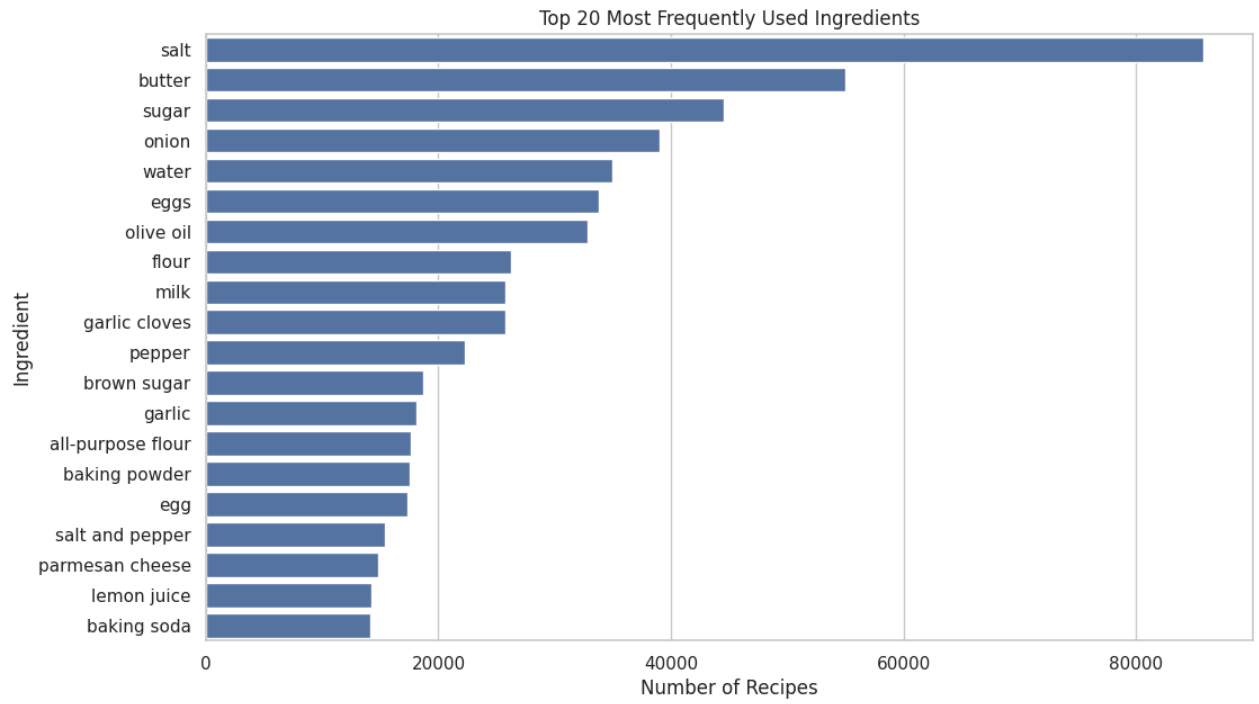
Next steps: Generate code with `ingredient_df` New interactive sheet

```
plt.figure(figsize=(12, 7))

sns.barplot(
    data=ingredient_df,
    x="Frequency",
    y="Ingredient"
)

plt.title("Top 20 Most Frequently Used Ingredients")
plt.xlabel("Number of Recipes")
plt.ylabel("Ingredient")

plt.show()
```




```
tag_counter = Counter()

for tags in recipes["tags"].dropna():
    try:
        tag_list = ast.literal_eval(tags)

        for tag in tag_list:
            tag_counter[tag.lower().strip()] += 1

    except (ValueError, SyntaxError):
        continue

top_20_tags = tag_counter.most_common(20)

tag_df = pd.DataFrame(
    top_20_tags,
    columns=["Tag", "Frequency"]
)

display(tag_df)
```

	Tag	Frequency
0	preparation	230546
1	time-to-make	225326
2	course	218148
3	main-ingredient	170446
4	dietary	165091
5	easy	126062
6	occasion	114145
7	cuisine	91165
8	low-in-something	85776
9	main-dish	71786
10	equipment	70436
11	60-minutes-or-less	69990
12	number-of-servings	58949
13	meat	56042
14	30-minutes-or-less	55077
15	vegetables	53814
16	taste-mood	52143
17	4-hours-or-less	49497
18	north-american	48479
19	3-steps-or-less	44933

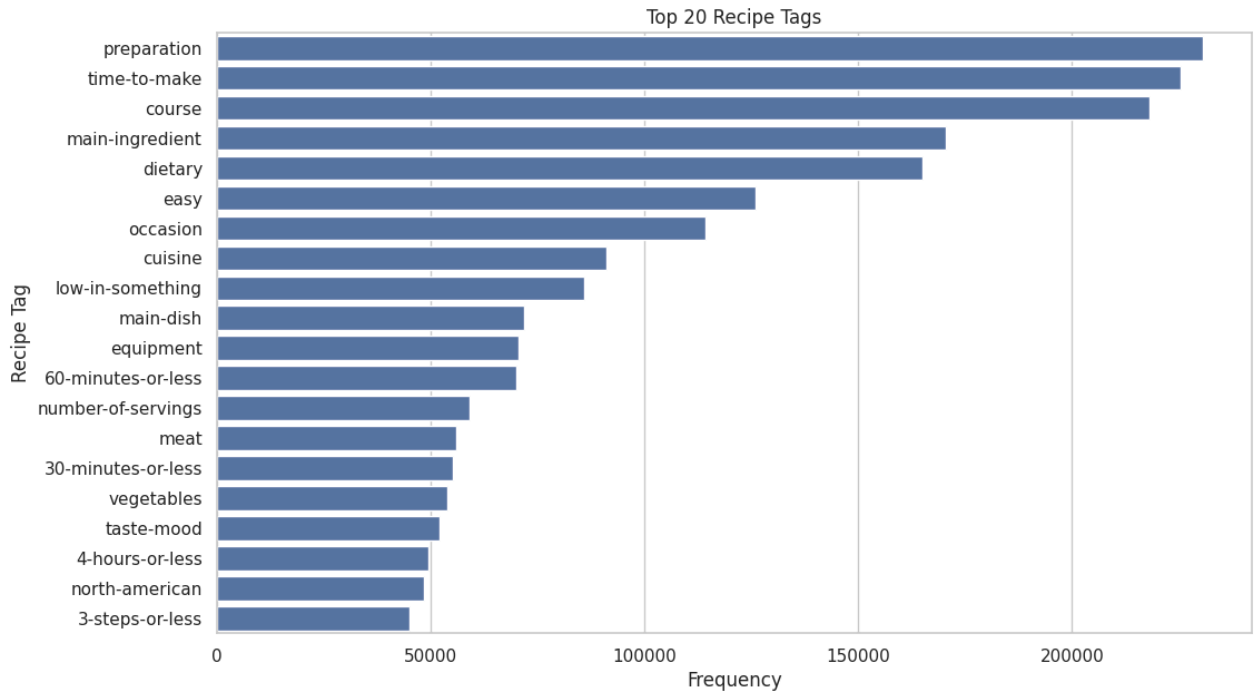
Next steps: Generate code with `tag_df` New interactive sheet

```
plt.figure(figsize=(12, 7))

sns.barplot(
    data=tag_df,
    x="Frequency",
    y="Tag"
)

plt.title("Top 20 Recipe Tags")
plt.xlabel("Frequency")
plt.ylabel("Recipe Tag")

plt.show()
```




```
display(recipes[["name", "nutrition"]].head(10))
```

	name	nutrition
0	arriba baked winter squash mexican style	[51.5, 0.0, 13.0, 0.0, 2.0, 0.0, 4.0]
1	a bit different breakfast pizza	[173.4, 18.0, 0.0, 17.0, 22.0, 35.0, 1.0]
2	all in the kitchen chili	[269.8, 22.0, 32.0, 48.0, 39.0, 27.0, 5.0]
3	alouette potatoes	[368.1, 17.0, 10.0, 2.0, 14.0, 8.0, 20.0]
4	amish tomato ketchup for canning	[352.9, 1.0, 337.0, 23.0, 3.0, 0.0, 28.0]
5	apple a day milk shake	[160.2, 10.0, 55.0, 3.0, 9.0, 20.0, 7.0]
6	aww marinated olives	[380.7, 53.0, 7.0, 24.0, 6.0, 24.0, 6.0]
7	backyard style barbecued ribs	[1109.5, 83.0, 378.0, 275.0, 96.0, 86.0, 36.0]
8	bananas 4 ice cream pie	[4270.8, 254.0, 1306.0, 111.0, 127.0, 431.0, 2...
9	beat this banana bread	[2669.3, 160.0, 976.0, 107.0, 62.0, 310.0, 138.0]

```
nutrition_columns = [
    "calories",
    "total_fat_pdv",
    "sugar_pdv",
    "sodium_pdv",
    "protein_pdv",
    "saturated_fat_pdv",
    "carbohydrates_pdv"
]

def parse_nutrition(value):

    try:
        values = ast.literal_eval(value)

        if isinstance(values, list) and len(values) == 7:
            return values

    except (ValueError, SyntaxError, TypeError):
        pass

    return [np.nan] * 7


nutrition_data = recipes["nutrition"].apply(parse_nutrition)

nutrition_df = pd.DataFrame(
    nutrition_data.tolist(),
```

```
    columns=nutrition_columns,
    index=recipes.index
)

display(nutrition_df.head())
```

	calories	total_fat_pdv	sugar_pdv	sodium_pdv	protein_pdv	saturated_fat_pdv	carbohydrates_pdv
0	51.5	0.0	13.0	0.0	2.0	0.0	4.0
1	173.4	18.0	0.0	17.0	22.0	35.0	1.0
2	269.8	22.0	32.0	48.0	39.0	27.0	5.0
3	368.1	17.0	10.0	2.0	14.0	8.0	20.0
4	352.9	1.0	337.0	23.0	3.0	0.0	28.0

```
recipes_eda = pd.concat(
    [recipes.copy(), nutrition_df],
    axis=1
)

display(
    recipes_eda[
        ["name", "calories", "protein_pdv",
         "total_fat_pdv", "carbohydrates_pdv"]
    ].head()
)
```

	name	calories	protein_pdv	total_fat_pdv	carbohydrates_pdv
0	arriba baked winter squash mexican style	51.5	2.0	0.0	4.0
1	a bit different breakfast pizza	173.4	22.0	18.0	1.0
2	all in the kitchen chili	269.8	39.0	22.0	5.0
3	alouette potatoes	368.1	14.0	17.0	20.0
4	amish tomato ketchup for canning	352.9	3.0	1.0	28.0

```
nutrition_df.describe()
```

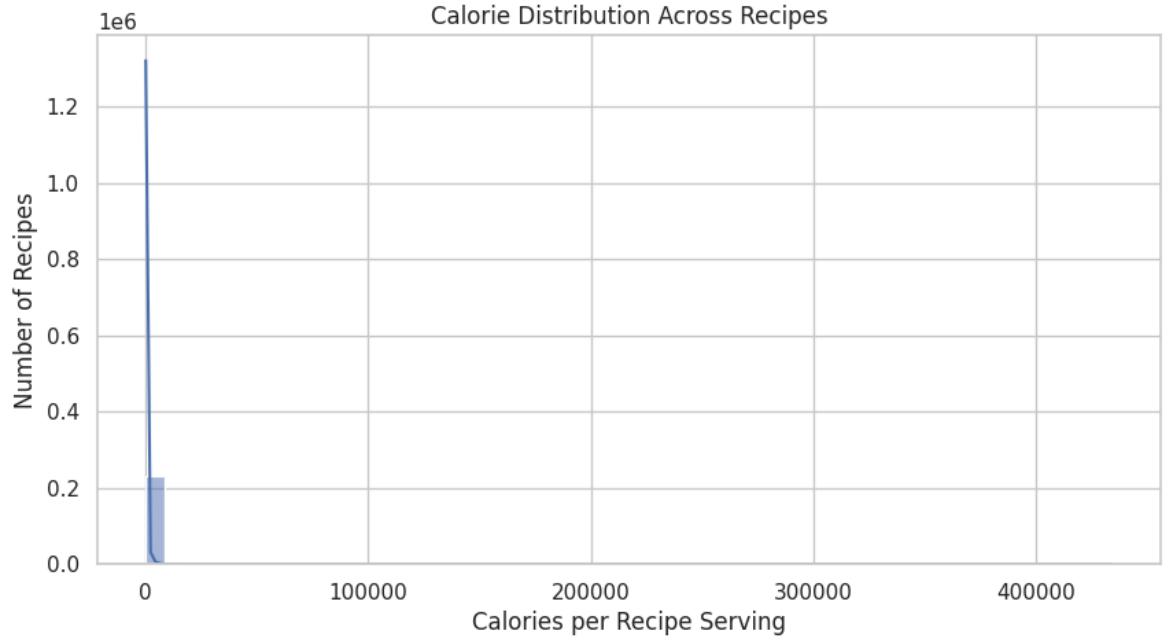
	calories	total_fat_pdv	sugar_pdv	sodium_pdv	protein_pdv	saturated_fat_pdv	carbohydrates_pdv
count	231637.000000	231637.00000	231637.000000	231637.000000	231637.00000	231637.000000	231637.000000
mean	473.942425	36.08070	84.296865	30.147485	34.68186	45.589150	15.560403
std	1189.711374	77.79884	800.080897	131.961589	58.47248	98.235758	81.824560
min	0.000000	0.00000	0.000000	0.000000	0.00000	0.000000	0.000000
25%	174.400000	8.00000	9.000000	5.000000	7.00000	7.000000	4.000000
50%	313.400000	20.00000	25.000000	14.000000	18.00000	23.000000	9.000000
75%	519.700000	41.00000	68.000000	33.000000	51.00000	52.000000	16.000000
max	434360.200000	17183.00000	362729.000000	29338.000000	6552.00000	10395.000000	36098.000000

```
plt.figure(figsize=(10, 5))

sns.histplot(
    nutrition_df["calories"].dropna(),
    bins=50,
    kde=True
)

plt.title("Calorie Distribution Across Recipes")
plt.xlabel("Calories per Recipe Serving")
plt.ylabel("Number of Recipes")

plt.show()
```

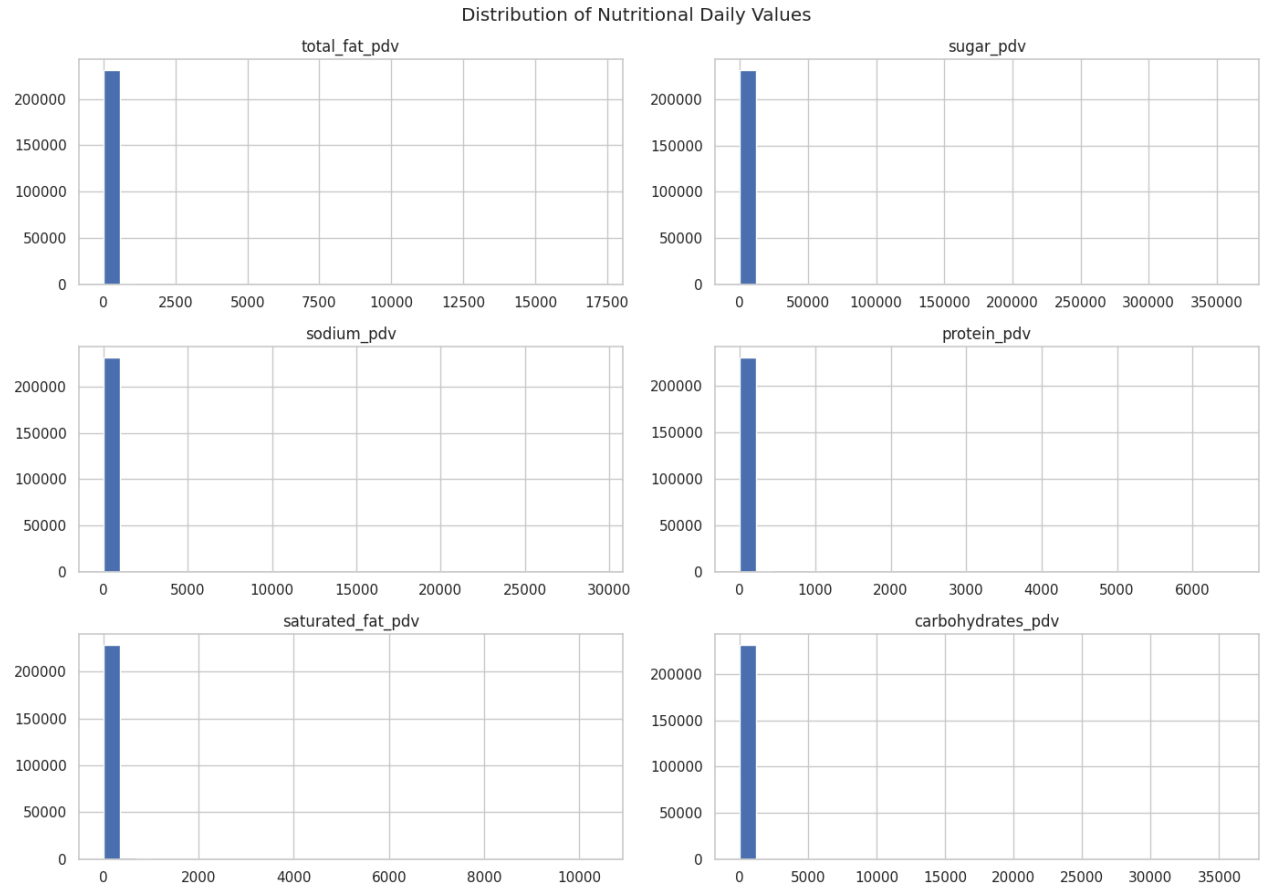



```
pdv_columns = [
    "total_fat_pdv",
    "sugar_pdv",
    "sodium_pdv",
    "protein_pdv",
    "saturated_fat_pdv",
    "carbohydrates_pdv"
]

nutrition_df[pdv_columns].hist(
    figsize=(14, 10),
    bins=30
)

plt.suptitle("Distribution of Nutritional Daily Values")

plt.tight_layout()
plt.show()
```




```
print("Total interactions:", len(interactions))
print("Unique users:", interactions["user_id"].nunique())
print("Unique recipes rated:", interactions["recipe_id"].nunique())

display(interactions.head())
```

```
Total interactions: 1132367
Unique users: 226570
Unique recipes rated: 231637
```

	user_id	recipe_id	date	rating	review
0	38094	40893	2003-02-17	4	Great with a salad. Cooked on top of stove for...
1	1293707	40893	2011-12-21	5	So simple, so delicious! Great for chilly fall...
2	8937	44394	2002-12-01	4	This worked very well and is EASY. I used not...
3	126440	85009	2010-02-27	5	I made the Mexican topping and took it to bunk...
4	57222	85009	2011-10-01	5	Made the cheddar bacon topping, adding a sprin...

```
print(interactions["rating"].value_counts().sort_index())
```

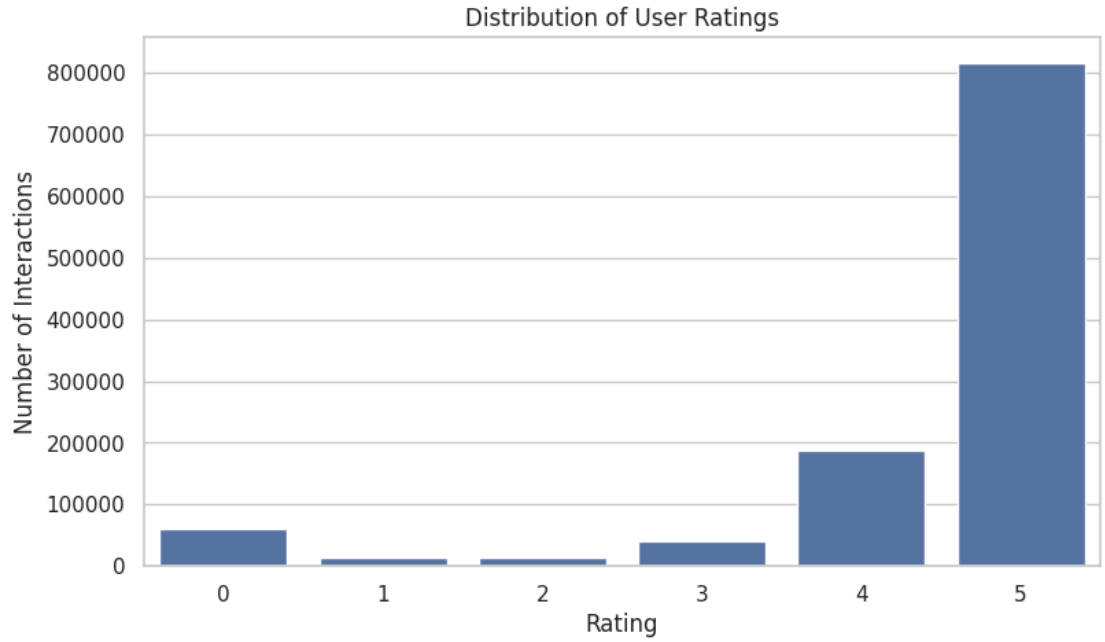
```
rating
0     60847
1     12818
2     14123
3     40855
4    187360
5    816364
Name: count, dtype: int64
```

```
plt.figure(figsize=(9, 5))

sns.countplot(
    data=interactions,
    x="rating"
)

plt.title("Distribution of User Ratings")
plt.xlabel("Rating")
plt.ylabel("Number of Interactions")
```

```
plt.show()
```




```
print("Average rating:", interactions["rating"].mean())
```

```
Average rating: 4.411016039852804
```

```
popular_recipes = (
    interactions
    .groupby("recipe_id")
    .agg(
        total_interactions=("rating", "count"),
        average_rating=("rating", "mean")
    )
    .sort_values(
        "total_interactions",
        ascending=False
    )
    .head(20)
)

display(popular_recipes)
```

recipe_id	total_interactions	average_rating
2886	1613	4.185989
27208	1601	4.288570
89204	1579	4.220393
39087	1448	4.541436
67256	1322	4.329047
54257	1305	4.213793
22782	1234	4.423015
32204	1220	4.525410
69173	997	4.708124
68955	904	4.398230
33919	877	4.374002
82102	855	4.605848
25885	847	4.440378
28148	802	4.699501
135350	786	4.249364
26110	770	4.254545
99476	762	4.279528
10744	731	4.496580
129926	730	4.309589
33671	727	4.466300

Next steps: Generate code with popular_recipes | New interactive sheet

```
popular_recipe_details = popular_recipes.merge(
    recipes[["id", "name"]],
    left_index=True,
    right_on="id",
    how="left"
)

display(
    popular_recipe_details[
        ["name", "total_interactions", "average_rating"]
    ]
)
```

	name	total_interactions	average_rating
22168	best banana bread	1613	4.185989
213826	to die for crock pot roast	1601	4.288570
66687	crock pot chicken with black beans cream cheese	1579	4.220393
62431	creamy cajun chicken pasta	1448	4.541436
22388	best ever banana cake with cream cheese frosting	1322	4.329047
229262	yes virginia there is a great meatloaf	1305	4.213793
115303	jo mama s world famous spaghetti	1234	4.423015
574	whatever floats your boat brownies	1220	4.525410
118161	kittencal s italian melt in your mouth meatballs	997	4.708124
114444	japanese mum s chicken	904	4.398230
62394	creamy burrito casserole	877	4.374002
118214	kittencal s moist cheddar garlic oven fried ch...	855	4.605848
16140	banana banana bread	847	4.440378
151142	oven fried chicken chimichangas	802	4.699501
82018	fannie farmer s classic baked macaroni cheese	786	4.249364
192283	southern buttermilk biscuits	770	4.254545
211450	the best easy beef and broccoli stir fry	762	4.279528
70791	delicious chicken pot pie	731	4.496580
187634	simply sour cream chicken enchiladas	730	4.309589
66630	crock pot whole chicken	727	4.466300

```
interactions["date"] = pd.to_datetime(
    interactions["date"],
    errors="coerce"
)

print(interactions["date"].describe())
```

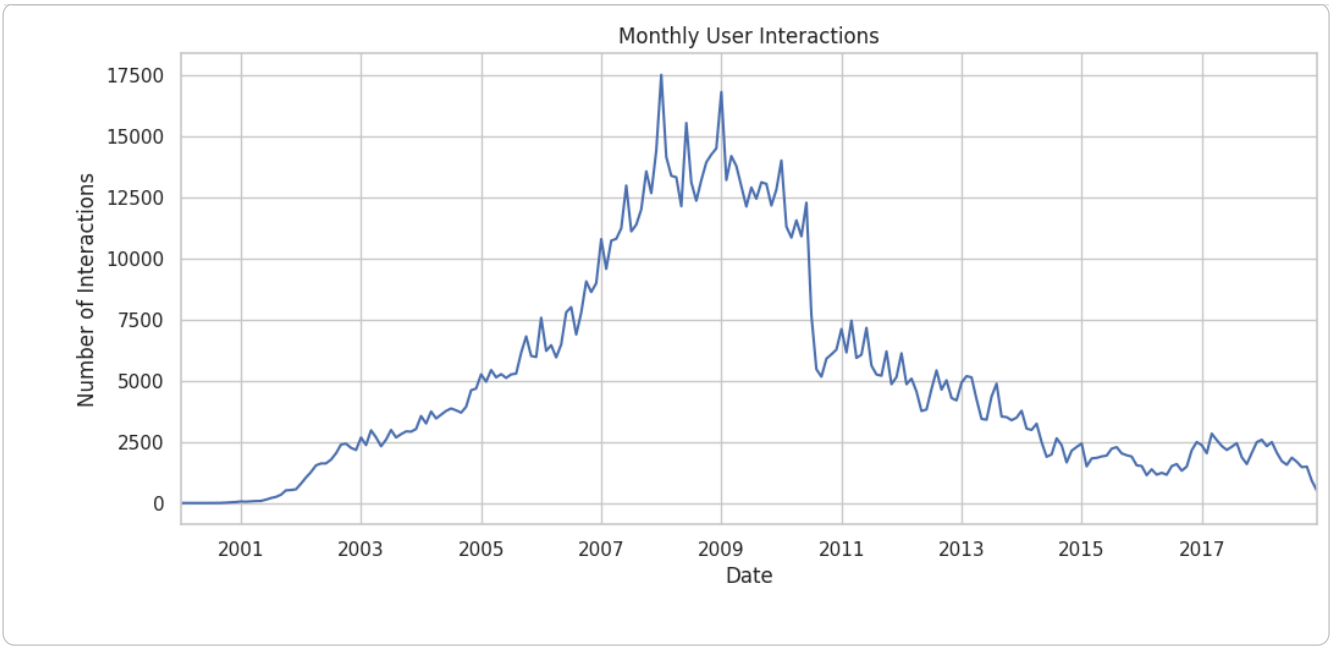
```
count                          1132367
mean     2009-05-13 02:57:38.777586944
min                2000-01-25 00:00:00
25%                2007-03-11 00:00:00
50%                2008-12-29 00:00:00
75%                2011-03-07 00:00:00
max                2018-12-20 00:00:00
Name: date, dtype: object
```

```
monthly_interactions = (
    interactions
    .dropna(subset=["date"])
    .set_index("date")
    .resample("ME")
    .size()
)

monthly_interactions.plot(figsize=(12, 5))

plt.title("Monthly User Interactions")
plt.xlabel("Date")
plt.ylabel("Number of Interactions")

plt.show()
```



Data Preprocessing and Feature Engineering.

```
import pandas as pd
import numpy as np
import ast
import matplotlib.pyplot as plt
import seaborn as sns
import re

from collections import Counter

print("Libraries imported successfully!")
```

```
Libraries imported successfully!
```

```
# Preserve the original datasets
recipes_raw = recipes.copy(deep=True)
interactions_raw = interactions.copy(deep=True)

print("Original datasets preserved!")
```

```
Original datasets preserved!
```

```
import kagglehub
import os

path = kagglehub.dataset_download(
    "shuyangli94/food-com-recipes-and-user-interactions"
)

recipes_raw = pd.read_csv(
    os.path.join(path, "RAW_recipes.csv")
)

interactions_raw = pd.read_csv(
    os.path.join(path, "RAW_interactions.csv")
)

print("Datasets loaded successfully!")
```

```
Using Colab cache for faster access to the 'food-com-recipes-and-user-interactions' dataset.
Datasets loaded successfully!
```

```
recipes = recipes_raw.copy(deep=True)
interactions = interactions_raw.copy(deep=True)

print("Working copies created!")
```

```
Working copies created!
```

```
print("RECIPES DATASET")
```

```
print("Shape:", recipes.shape)

print("\nINTERACTIONS DATASET")
print("Shape:", interactions.shape)
```

```
RECIPES DATASET
Shape: (231637, 12)

INTERACTIONS DATASET
Shape: (1132367, 5)
```

```
print("Recipes unchanged:",
      recipes_raw.equals(recipes))

print("Interactions unchanged:",
      interactions_raw.equals(interactions))
```

```
Recipes unchanged: True
Interactions unchanged: True
```

```
print(recipes["minutes"].describe())
```

```
count    2.316370e+05
mean     9.398546e+03
std      4.461963e+06
min      0.000000e+00
25%      2.000000e+01
50%      4.000000e+01
75%      6.500000e+01
max      2.147484e+09
Name: minutes, dtype: float64
```

```
print("Recipes taking more than 1 day:")

print(
    recipes.loc[
        recipes["minutes"] > 1440,
        ["id", "name", "minutes"]
    ].sort_values(
        by="minutes",
        ascending=False
    ).head(20)
)
```

```
Recipes taking more than 1 day:
            id                                              name     minutes
144074  261647                             no bake granola balls  2147483647
109624  447963                           how to preserve a husband     1051200
106563  291571                              homemade fruit liquers      288000
38116   216215    celtic druid s honey mead   meade   metheglin      259260
106963  425681                                 homemade vanilla      259205
155044   98912                                     peach brandy      216015
5497     70551                                  angelica liqueur      201610
177906   12931                                           rumtopf      187200
170999  172114                                raspberry liqueur      172810
53721   153634                                 cinnamon extract      172800
102299    2714                                 hard apple cider      167040
171791  236340  recipe for elephant stew  from griot s cookbook      146880
41471    13345                                   cherry liqueur      144015
145524  242032                                         nut wine      132495
85835   240595                                       framboise      129630
220566  184528                                  vanilla extract      129620
26124   236281                                blueberry liqueur      129615
201041  236274                               strawberry liqueur      129615
6545    236379                            apple cinnamon liqueur      129615
186635  193914                                     simple brandy      129610
```

```
time_ranges = {
    "Zero or negative": (recipes["minutes"] <= 0).sum(),
    "1–15 minutes": recipes["minutes"].between(1, 15).sum(),
    "16–30 minutes": recipes["minutes"].between(16, 30).sum(),
    "31–60 minutes": recipes["minutes"].between(31, 60).sum(),
    "61–120 minutes": recipes["minutes"].between(61, 120).sum(),
    "121–1440 minutes": recipes["minutes"].between(121, 1440).sum(),
    "More than 1440 minutes": (recipes["minutes"] > 1440).sum()
}

for category, count in time_ranges.items():
    print(f"{category}: {count}")
```

```
Zero or negative: 1094
1–15 minutes: 42828
16–30 minutes: 55131
31–60 minutes: 70077
61–120 minutes: 36683
121–1440 minutes: 23824
More than 1440 minutes: 2000
```

```
zero_count = (recipes["minutes"] == 0).sum()
negative_count = (recipes["minutes"] < 0).sum()

print("Zero preparation times:", zero_count)
print("Negative preparation times:", negative_count)
```

```
Zero preparation times: 1094
Negative preparation times: 0
```

```
invalid_time_recipes = recipes[
    recipes["minutes"] <= 0
][["id", "name", "minutes", "ingredients", "tags"]]

print("Total records:", len(invalid_time_recipes))

display(invalid_time_recipes.head(20))
```

```
Total records: 1094
```

	id	name	minutes	ingredients	tags
5	5289	apple a day milk shake	0	['milk', 'vanilla ice cream', 'frozen apple ju...	['15-minutes-or-less', 'time-to-make', 'course...
2451	1712	acorn magic delights	0	['butter', 'brown sugar', 'pecans', 'all-purpo...	['15-minutes-or-less', 'time-to-make', 'course...
3079	4880	albanian byrek	0	['flour', 'water', 'oil', 'vinegar', 'salt', '...	['15-minutes-or-less', 'time-to-make', 'course...
3193	3258	alfredo sauce with pasta	0	['butter', 'heavy cream', 'parmesan cheese', '...	['15-minutes-or-less', 'time-to-make', 'course...
3259	2284	alice s doughnuts	0	['eggs', 'sugar', 'milk', 'shortening', 'vanil...	['15-minutes-or-less', 'time-to-make', 'course...
3417	6648	all bran seed loaf	0	['whole wheat flour', 'flour', 'baking powder'...	['15-minutes-or-less', 'time-to-make', 'course...
3440	7009	all in one tuna casserole	0	['onion soup mix', 'milk', 'frozen peas and ca...	['15-minutes-or-less', 'time-to-make', 'course...
3571	4478	almond bread	0	['egg whites', 'flour', 'sugar', 'whole almonds']	['15-minutes-or-less', 'time-to-make', 'course...
4889	7550	americana key lime pie	0	['unflavored gelatin', 'egg whites', 'sugar', ...	['15-minutes-or-less', 'time-to-make', 'course...
4918	1420	amish biscuits	0	['all-purpose flour', 'baking soda', 'salt', '...	['15-minutes-or-less', 'time-to-make', 'course...
5015	3689	amish pot roast	0	['swiss steak', 'vegetable oil', 'soy sauce', ...	['15-minutes-or-less', 'time-to-make', 'cuisin...
5153	43301	an mochi 1	0	['mochiko sweet rice flour', 'sugar', 'water',...	['15-minutes-or-less', 'time-to-make', 'course...
5266	6511	andouille in comforting barbecue sauce	0	['onions', 'celery', 'bell pepper', 'parsley',...	['weeknight', '15-minutes-or-less', 'time-to-m...
5548	5179	anise biscotti	0	['all-purpose flour', 'sugar', 'anise seed', '...	['15-minutes-or-less', 'time-to-make', 'course...
5565	5178	anise almond biscotti	0	['almonds', 'flour', 'baking powder', 'salt', ...	['15-minutes-or-less', 'time-to-make', 'course...
5725	7590	another bean salad	0	['sugar', 'salt', 'vinegar', 'green beans', 'y...	['15-minutes-or-less', 'time-to-make', 'course...
5958	5197	anytime cheese ball	0	['philadelphia cream cheese', 'sharp cheese', ...	['15-minutes-or-less', 'time-to-make', 'main-i...
6130	2684	apple and cheddar corn muffins	0	['flour', 'yellow cornmeal', 'baking powder', ...	['15-minutes-or-less', 'time-to-make', 'course...
6436	3667	apple cheese crisp	0	['butter', 'brown sugar', 'oats', 'flour', 'ci...	['15-minutes-or-less', 'time-to-make', 'course...
6437	3838	apple cheese crisp ii	0	['apples', 'sugar', 'brown sugar', 'lemon, rin...	['15-minutes-or-less', 'time-to-make', 'course...

```
extreme_recipes = recipes[
    recipes["minutes"] > 1440
][["id", "name", "minutes"]].sort_values(
    by="minutes",
    ascending=False
)

display(extreme_recipes.head(30))
```

	id	name	minutes
144074	261647	no bake granola balls	2147483647
109624	447963	how to preserve a husband	1051200
106563	291571	homemade fruit liquers	288000
38116	216215	celtic druid s honey mead meade metheglin	259260
106963	425681	homemade vanilla	259205
155044	98912	peach brandy	216015
5497	70551	angelica liqueur	201610
177906	12931	rumtopf	187200
170999	172114	raspberry liqueur	172810
53721	153634	cinnamon extract	172800
102299	2714	hard apple cider	167040
171791	236340	recipe for elephant stew from griot s cookbook	146880
41471	13345	cherry liqueur	144015
145524	242032	nut wine	132495
85835	240595	framboise	129630
220566	184528	vanilla extract	129620
26124	236281	blueberry liqueur	129615
201041	236274	strawberry liqueur	129615
6545	236379	apple cinnamon liqueur	129615
186635	193914	simple brandy	129610
106964	463624	homemade vanilla extract	129600
124353	13264	limoncello lemoncella	115230
69472	165697	dandelion wine	87842
155096	435928	peach cordial	86415
188486	274757	sloe gin	86415
24807	147401	blackberry brandy	86410
27153	95215	bouillon bbq onions	86410
115341	98818	joanne s sweet and hot pickles	86407
178202	105417	ruth s homemade vanilla extract	86405
89732	34175	garden hot sauce	86400

```
# Create a separate cleaned column
recipes["minutes_clean"] = recipes["minutes"].copy()

# Mark zero and negative values as missing
recipes.loc[
    recipes["minutes_clean"] <= 0,
    "minutes_clean"
] = np.nan

# Mark the known extreme invalid value as missing
recipes.loc[
    recipes["minutes_clean"] == 2147483647,
    "minutes_clean"
] = np.nan

print("Preparation time cleaning completed!")
```

```
Preparation time cleaning completed!
```

```
print("Original preparation time statistics:")
print(recipes["minutes"].describe())

print("\nCleaned preparation time statistics:")
print(recipes["minutes_clean"].describe())

print("\nMissing values in cleaned preparation time:",
      recipes["minutes_clean"].isnull().sum())
```

```
Original preparation time statistics:
count    2.316370e+05
mean     9.398546e+03
std      4.461963e+06
min      0.000000e+00
25%      2.000000e+01
50%      4.000000e+01
75%      6.500000e+01
max      2.147484e+09
Name: minutes, dtype: float64

Cleaned preparation time statistics:
count    2.305420e+05
mean     1.282515e+02
std      2.953313e+03
min      1.000000e+00
25%      2.000000e+01
50%      4.000000e+01
75%      6.500000e+01
max      1.051200e+06
Name: minutes_clean, dtype: float64

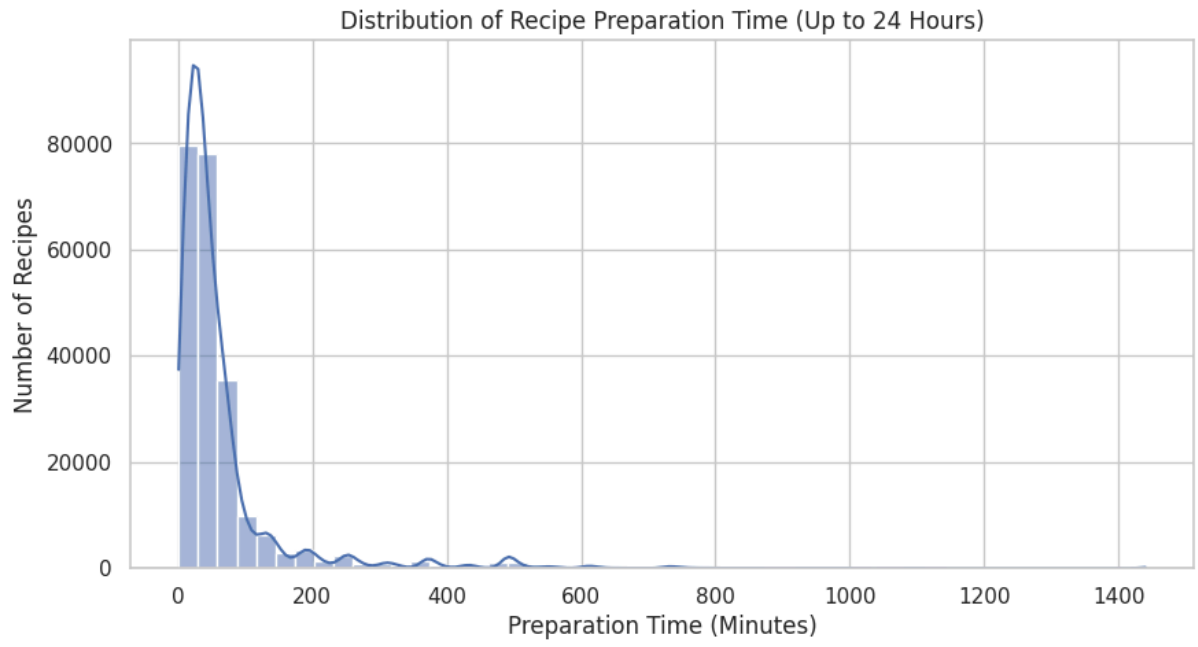
Missing values in cleaned preparation time: 1095
```

```
plt.figure(figsize=(10, 5))

sns.histplot(
    recipes.loc[
        recipes["minutes_clean"].between(1, 1440),
        "minutes_clean"
    ],
    bins=50,
    kde=True
)

plt.title("Distribution of Recipe Preparation Time (Up to 24 Hours)")
plt.xlabel("Preparation Time (Minutes)")
plt.ylabel("Number of Recipes")

plt.show()
```




```
def categorize_time(minutes):
```

```
    if pd.isna(minutes):
        return "Unknown"

    elif minutes <= 15:
        return "Quick"

    elif minutes <= 30:
        return "Short"

    elif minutes <= 60:
        return "Medium"

    else:
        return "Long"


recipes["time_category"] = recipes["minutes_clean"].apply(
    categorize_time
)

print(recipes["time_category"].value_counts())
```

```
time_category
Medium     70077
Long       62506
Short      55131
Quick      42828
Unknown     1095
Name: count, dtype: int64
```

```
print("First 10 nutrition records:")

for value in recipes["nutrition"].head(10):
    print(value)
```

```
First 10 nutrition records:
[51.5, 0.0, 13.0, 0.0, 2.0, 0.0, 4.0]
[173.4, 18.0, 0.0, 17.0, 22.0, 35.0, 1.0]
[269.8, 22.0, 32.0, 48.0, 39.0, 27.0, 5.0]
[368.1, 17.0, 10.0, 2.0, 14.0, 8.0, 20.0]
[352.9, 1.0, 337.0, 23.0, 3.0, 0.0, 28.0]
[160.2, 10.0, 55.0, 3.0, 9.0, 20.0, 7.0]
[380.7, 53.0, 7.0, 24.0, 6.0, 24.0, 6.0]
[1109.5, 83.0, 378.0, 275.0, 96.0, 86.0, 36.0]
[4270.8, 254.0, 1306.0, 111.0, 127.0, 431.0, 220.0]
[2669.3, 160.0, 976.0, 107.0, 62.0, 310.0, 138.0]
```

```
nutrition_columns = [
    "calories",
    "total_fat_pdv",
    "sugar_pdv",
    "sodium_pdv",
    "protein_pdv",
    "saturated_fat_pdv",
    "carbohydrates_pdv"
]

def parse_nutrition(value):

    try:
        parsed = ast.literal_eval(value)

        if isinstance(parsed, list) and len(parsed) == 7:
            return parsed

        return [np.nan] * 7

    except (ValueError, SyntaxError, TypeError):
        return [np.nan] * 7


nutrition_values = recipes["nutrition"].apply(parse_nutrition)

nutrition_df = pd.DataFrame(
    nutrition_values.tolist(),
    columns=nutrition_columns,
    index=recipes.index
)

# Add nutritional features to the working recipe dataset
recipes[nutrition_columns] = nutrition_df
```

```
print("Nutrition parsing completed!")

display(recipes[
    ["name"] + nutrition_columns
].head())
```

```
Nutrition parsing completed!
```

	name	calories	total_fat_pdv	sugar_pdv	sodium_pdv	protein_pdv	saturated_fat_pdv	carbohydrates_pdv
0	arriba baked winter squash mexican style	51.5	0.0	13.0	0.0	2.0	0.0	4.0
1	a bit different breakfast pizza	173.4	18.0	0.0	17.0	22.0	35.0	1.0
2	all in the kitchen chili	269.8	22.0	32.0	48.0	39.0	27.0	5.0
3	alouette	368.1	17.0	10.0	2.0	14.0	8.0	20.0

```
print("NUTRITION DATA TYPES")

print(recipes[nutrition_columns].dtypes)

print("\nMISSING NUTRITION VALUES")

print(recipes[nutrition_columns].isnull().sum())
```

```
NUTRITION DATA TYPES
calories             float64
total_fat_pdv        float64
sugar_pdv            float64
sodium_pdv           float64
protein_pdv          float64
saturated_fat_pdv    float64
carbohydrates_pdv    float64
dtype: object

MISSING NUTRITION VALUES
calories             0
total_fat_pdv        0
sugar_pdv            0
sodium_pdv           0
protein_pdv          0
saturated_fat_pdv    0
carbohydrates_pdv    0
dtype: int64
```

```
nutrition_stats = recipes[nutrition_columns].describe().T

display(nutrition_stats)
```

	count	mean	std	min	25%	50%	75%	max
calories	231637.0	473.942425	1189.711374	0.0	174.4	313.4	519.7	434360.2
total_fat_pdv	231637.0	36.080700	77.798840	0.0	8.0	20.0	41.0	17183.0
sugar_pdv	231637.0	84.296865	800.080897	0.0	9.0	25.0	68.0	362729.0
sodium_pdv	231637.0	30.147485	131.961589	0.0	5.0	14.0	33.0	29338.0
protein_pdv	231637.0	34.681860	58.472480	0.0	7.0	18.0	51.0	6552.0
saturated_fat_pdv	231637.0	45.589150	98.235758	0.0	7.0	23.0	52.0	10395.0
carbohydrates_pdv	231637.0	15.560403	81.824560	0.0	4.0	9.0	16.0	36098.0

Next steps: Generate code with `nutrition_stats` New interactive sheet

```
for column in nutrition_columns:

    print("\n" + "=" * 60)
    print("FEATURE:", column)
    print("=" * 60)

    display(
        recipes[["id", "name", column]]
        .sort_values(by=column, ascending=False)
```

```
            .head(10)
        )
```

```
============================================================
FEATURE: calories
============================================================
```

	id	name	calories
209453	193435	tennessee moonshine	434360.2
70467	227470	deep fried prime rib	101614.7
165504	392286	powdered hot cocoa mix	45609.0
137181	82848	mmmmmmmmilky way cake	44239.8
34023	18330	caledonian wedding cake	43924.6
141997	75591	my mercy strawberry cake	42951.6
78689	48363	easy easy friendship cake	40159.2
182816	59776	scott hibb s amazing whiskey grilled baby back...	38680.1
199615	93481	steph s pork riblets	38662.3
139058	319406	moonshine easy	36188.8

```
============================================================
FEATURE: total_fat_pdv
============================================================
```

	id	name	total_fat_pdv
70467	227470	deep fried prime rib	17183.0
199615	93481	steph s pork riblets	4331.0
182816	59776	scott hibb s amazing whiskey grilled baby back...	4317.0
56433	500039	coffee glazed doughnuts	3464.0
165504	392286	powdered hot cocoa mix	3379.0
84023	260961	finger licking sticky juicy ribs	3229.0
164719	32152	potato crumble	3105.0
102405	256795	harold s florentine salad dressing	3072.0
134453	232495	mexican ojarascas cookies	3021.0
219494	313271	ultimate coconut cake ii	2899.0

```
============================================================
FEATURE: sugar_pdv
============================================================
```

	id	name	sugar_pdv
209453	193435	tennessee moonshine	362729.0
139058	319406	moonshine easy	30260.0
34023	18330	caledonian wedding cake	18943.0
107018	386564	homesteader s fireweed honey	18127.0
137181	82848	mmmmmmmmilky way cake	17306.0
165504	392286	powdered hot cocoa mix	16901.0
95612	535853	granny jones secret salty sweet biscuit recipe	14495.0
105817	533590	hocus pocus cottage cake	13573.0
931	17490	14 day unsealed pickles	12914.0
187719	46942	sinfully delicious hot chocolate mix	11042.0

```
============================================================
FEATURE: sodium_pdv
============================================================
```

	id	name	sodium_pdv
95612	535853	granny jones secret salty sweet biscuit recipe	29338.0
137168	223287	mmmm salt encrusted baked fish	14664.0

```
high_calorie_recipes = recipes.sort_values(
    by="calories",
    ascending=False
).head(15)
```

```
display(
    high_calorie_recipes[
        [
            "id",
            "name",
            "minutes_clean",
            "n_ingredients",
            "calories",
            "total_fat_pdv",
            "sugar_pdv",
            "sodium_pdv",
            "protein_pdv",
            "saturated_fat_pdv",
            "carbohydrates_pdv",
            "ingredients"
        ]
    ]
)
```

165504	392286	powdered hot cocoa mix	4956.0
95462	101471	grandmag s baked christmas ham	4213.0
209453	193435	tennessee moonshine	3961.0
4252	338356	alternate honey barbecue sauce with riblets a...	3605.0
158521	139292	perfect turkey	3351.0
107215	250949	honey barbecue riblets applebee s	3268.0
14283	77876	baked glazed ham	3156.0

```
============================================================
FEATURE: saturated_fat_pdv
============================================================
```

	id	name	saturated_fat_pdv
70467	227470	deep fried prime rib	10395.0
165504	392286	powdered hot cocoa mix	6875.0
164719	32152	potato crumble	6385.0
219494	313271	ultimate coconut cake ii	6269.0
199615	93481	steph s pork riblets	4969.0
182816	59776	scott hibb s amazing whiskey grilled baby back...	4960.0
34023	18330	caledonian wedding cake	4941.0
212993	394520	this is the real hooters wing sauce for the ho...	4610.0
55796	45389	coconut macaroons	4209.0
154459	31130	patience coconut cake	3730.0

```
============================================================
FEATURE: carbohydrates_pdv
============================================================
```

	id	name	carbohydrates_pdv
209453	193435	tennessee moonshine	36098.0
139058	319406	moonshine easy	3007.0
137181	82848	mmmmmmmmilky way cake	2565.0
141997	75591	my mercy strawberry cake	2315.0
78689	48363	easy easy friendship cake	2275.0
34023	18330	caledonian wedding cake	2233.0
105817	533590	hocus pocus cottage cake	1554.0
107018	386564	homesteader s fireweed honey	1511.0
165504	392286	powdered hot cocoa mix	1453.0
95612	535853	granny jones secret salty sweet biscuit recipe	1440.0

	id	name	minutes_clean	n_ingredients	calories	total_fat_pdv	sugar_pdv	sodium_pdv	protein_pd
209453	193435	tennessee moonshine	20.0	5	434360.2	1293.0	362729.0	1324.0	3961
70467	227470	deep fried prime rib	100.0	6	101614.7	17183.0	0.0	230.0	1271
165504	392286	powdered hot cocoa mix	10.0	4	45609.0	3379.0	16901.0	1478.0	4356
137181	82848	mmmmmmmmilky way cake	60.0	6	44239.8	2036.0	17306.0	2705.0	936
34023	18330	caledonian wedding cake	630.0	17	43924.6	2556.0	18943.0	841.0	1103
141997	75591	my mercy strawberry cake	60.0	9	42951.6	2343.0	1544.0	2405.0	886
78689	48363	easy easy friendship cake	75.0	10	40159.2	1922.0	1024.0	2386.0	810
182816	59776	scott hibb s amazing whiskey grilled baby back...	180.0	18	38680.1	4317.0	185.0	686.0	6540
199615	93481	steph s pork riblets	70.0	7	38662.3	4331.0	77.0	809.0	6552
139058	319406	moonshine easy	7200.0	4	36188.8	106.0	30260.0	80.0	329
183541	148446	seasoned goldfish crackers	30.0	7	30933.4	2406.0	68.0	2538.0	1237
84023	260961	finger licking sticky juicy ribs	50.0	10	29088.7	3229.0	172.0	555.0	4909
219494	313271	ultimate coconut cake ii	120.0	16	28930.2	2899.0	9245.0	307.0	360
105817	533590	hocus pocus cottage cake	3000.0	27	26604.4	1321.0	13573.0	595.0	446
187719	46942	sinfully delicious hot chocolate mix	5.0	4	25712.6	1715.0	11042.0	684.0	2238

```
print("ZERO VALUES")

display(
```

```
    (recipes[nutrition_columns] == 0).sum()
)

print("\nNEGATIVE VALUES")

display(
    (recipes[nutrition_columns] < 0).sum()
```

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